

Deep Generative Models

Lecture 5

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Recap of Previous Lecture

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ and $\epsilon \sim p(\epsilon)$.
2. Reparametrize $\mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)$.
3. Compute the ELBO:

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) \approx \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})).$$

4. Update ϕ, θ via stochastic gradient ascent.

Sampling

1. Sample $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$.
2. Generate $\mathbf{x} \sim p_{\theta}(\mathbf{x}|\mathbf{z})$.

Note: The encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$ isn't needed during generation.

Recap of Previous Lecture

Theorem

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \parallel p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}].$$

ELBO Surgery

$$\frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi,\theta}(\mathbf{x}_i) = \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} - \underbrace{\mathbb{I}_q[\mathbf{x}, \mathbf{z}]}_{\text{MI}} - \underbrace{D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z}))}_{\text{Marginal KL}}$$

Optimal Prior

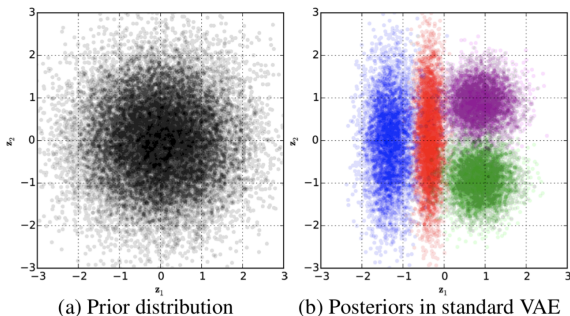
$$D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) = 0 \Leftrightarrow p(\mathbf{z}) = q_{\text{agg}}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i).$$

Thus, the optimal prior distribution $p(\mathbf{z})$ is the aggregated variational posterior $q_{\text{agg},\phi}(\mathbf{z})$.

Hoffman M. D., Johnson M. J. *ELBO surgery: yet another way to carve up the variational evidence lower bound*, 2016

Recap of Previous Lecture

- ▶ **Prior mismatch:** The unimodal encoder $q_\phi(\mathbf{z}|\mathbf{x})$ yields $q_{\text{agg},\phi}(\mathbf{z})$ that often does not match the Gaussian prior $p(\mathbf{z})$.
- ▶ **Blurriness from averaging:** With Gaussian decoder, ELBO minimization gives $\mu^*(\mathbf{z}) = \mathbb{E}_{q_\phi(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$. If distinct inputs $\mathbf{x} \neq \mathbf{x}'$ map to *overlapping* latent regions, the decoder averages over unrelated data.



Recap of Previous Lecture

Assumptions

- ▶ Let $c \sim \text{Cat}(\boldsymbol{\pi})$, where

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^K \pi_k = 1.$$

- ▶ Suppose the VAE employs a discrete latent code c , with prior $p(c) = \text{Uniform}\{1, \dots, K\}$.

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x}|c) - D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) \rightarrow \max_{\phi, \theta}.$$

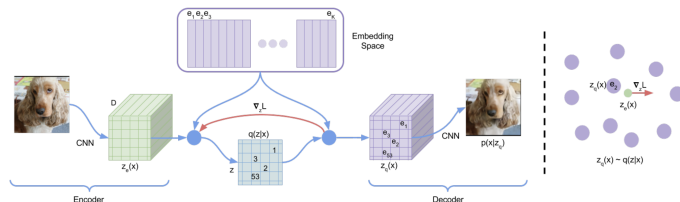
$$D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) = -H(q_{\phi}(c|\mathbf{x})) + \log K.$$

Vector Quantization

Define the codebook $\{\mathbf{e}_k\}_{k=1}^K$, where $\mathbf{e}_k \in \mathbb{R}^L$ and K is the size of the dictionary.

$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg \min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

Recap of Previous Lecture



Deterministic Variational Posterior

$$q_{\phi}(c_{ij} = k^* | \mathbf{x}) = \begin{cases} 1, & \text{if } k^* = \arg \min_k \|[z_e]_{ij} - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x} | \mathbf{e}_c) - \log K = \log p_{\theta}(\mathbf{x} | \mathbf{z}_q) - \log K.$$

Straight-Through Gradient Estimation

$$\frac{\partial \log p(\mathbf{x} | \mathbf{z}_q, \theta)}{\partial \phi} = \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \phi} \approx \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \phi}$$

Recap of Previous Lecture

Likelihood-Free Learning

- ▶ Likelihood isn't a perfect metric for generative models.
- ▶ Likelihood may be intractable.

Imagine we have two sets of samples:

- ▶ $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ – real samples;
- ▶ $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\theta}(\mathbf{x})$ – generated (fake) samples.

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\theta}(\mathbf{x}))$$

Assumption

The generative distribution $p_{\theta}(\mathbf{x})$ matches the true distribution $p_{\text{data}}(\mathbf{x})$ if we can't distinguish between them using a discriminative model $p(y|\mathbf{x})$.

- ▶ **Generator:** a generative model $\mathbf{x} = \mathbf{G}(\mathbf{z})$ that produces more realistic samples.
- ▶ **Discriminator:** a classifier $D(\mathbf{x}) \in [0, 1]$ distinguishing real from generated samples.

Outline

1. Generative Adversarial Networks (GAN)
2. Wasserstein Distance
3. Wasserstein GAN (WGAN)
4. Evaluation of Likelihood-Free Models
 - Frechet Inception Distance (FID)
 - Precision-Recall
 - CLIP Score
 - Human Eval

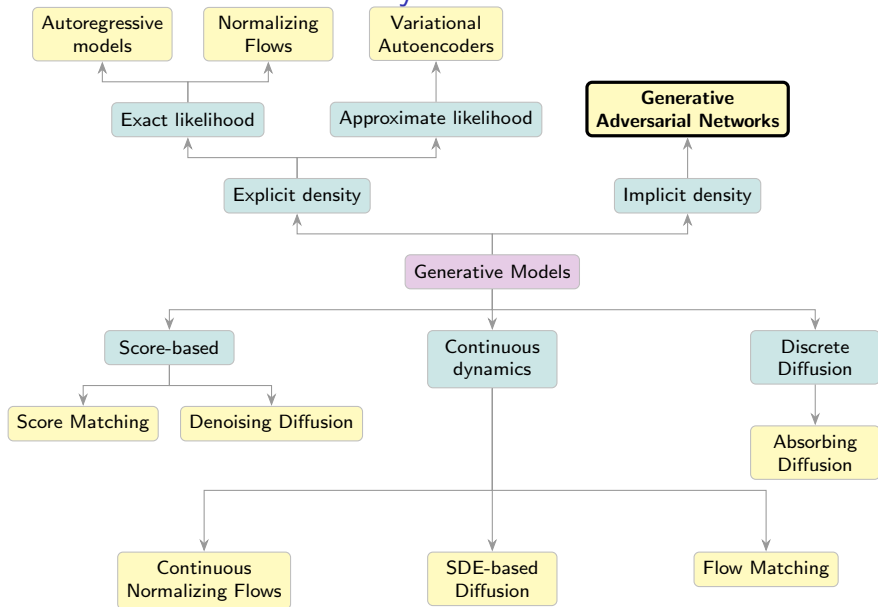
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Generative Models Taxonomy



GAN Optimality

Theorem

The minimax game

$$\min_G \max_D \underbrace{\left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]}_{V(G,D)}$$

achieves its global optimum when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

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Proof (Fixed G)

$$V(G, D) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x}))$$

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$$\begin{aligned} V(G, D) &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \\ &= \int \underbrace{[p_{\text{data}}(\mathbf{x}) \log D(\mathbf{x}) + p_{\theta}(\mathbf{x}) \log(1 - D(\mathbf{x}))]}_{y(D)} d\mathbf{x} \end{aligned}$$

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$$\frac{dy(D)}{dD} = \frac{p_{\text{data}}(\mathbf{x})}{D(\mathbf{x})} - \frac{p_{\theta}(\mathbf{x})}{1 - D(\mathbf{x})} = 0 \quad \Rightarrow \quad D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})}$$

GAN Optimality

Proof Continued (Fixed $D = D^*$)

$$V(G, D^*) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log \left(\frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log \left(\frac{p_{\theta}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right)$$

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Jensen-Shannon Divergence (Symmetric KL Divergence)

$$D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) = \frac{1}{2} [D_{\text{KL}}(p_{\text{data}}(\mathbf{x}) \parallel \star) + D_{\text{KL}}(p_{\theta}(\mathbf{x}) \parallel \star)]$$

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This can be regarded as a proper distance metric!

$$V(G^*, D^*) = -2 \log 2, \quad p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x}), \quad D^*(\mathbf{x}) = 0.5.$$

GAN Optimality

Theorem

The following minimax game

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achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

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Reality

- ▶ Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- ▶ Generator and discriminator losses typically oscillate during GAN training.

GAN Training

Assume both generator and discriminator are parametric models:
 $D_\phi(\mathbf{x})$ and $\mathbf{G}_\theta(\mathbf{z})$.

Objective

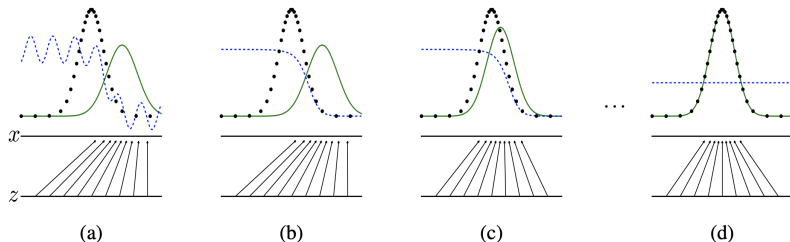
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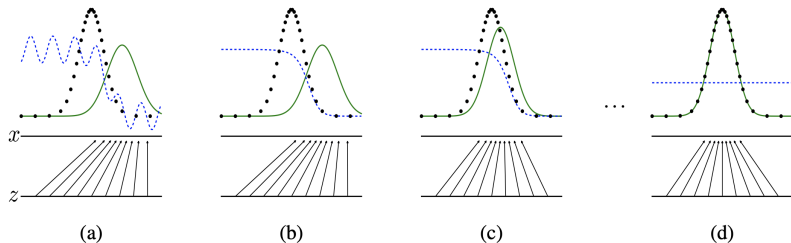


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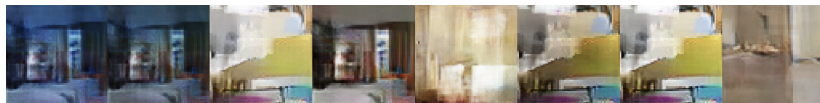
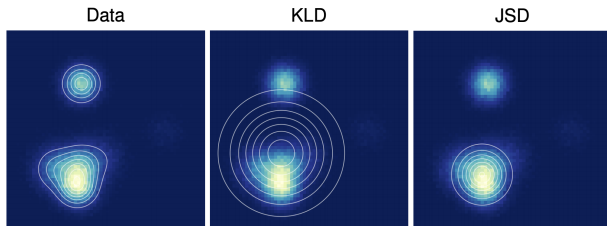
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- ▶ $\mathbf{z} \sim p(\mathbf{z})$ is a latent variable.
- ▶ $p_\theta(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_\theta(\mathbf{z}))$ serves as a deterministic decoder (like normalizing flows).
- ▶ There is no encoder present.

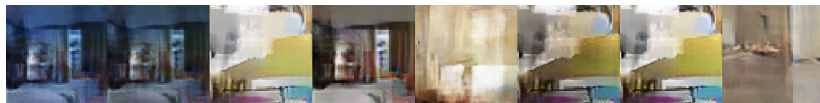
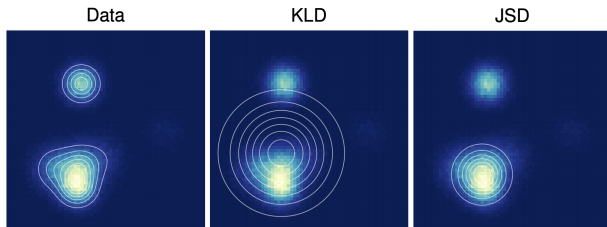
Mode Collapse

Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.



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Numerous methods have been proposed to tackle mode collapse: changing architectures, adding regularization terms, injecting noise.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014

Metz L. et al. Unrolled Generative Adversarial Networks, 2016

Jensen-Shannon vs Kullback-Leibler Divergences

- ▶ $p_{\text{data}}(\mathbf{x})$ is a fixed mixture of two Gaussians.
- ▶ $p(\mathbf{x}|\mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$.

Mode Covering vs. Mode Seeking

$$D_{\text{KL}}(\pi \parallel p) = \int \pi(\mathbf{x}) \log \frac{\pi(\mathbf{x})}{p(\mathbf{x})} d\mathbf{x}, \quad D_{\text{KL}}(p \parallel \pi) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$$

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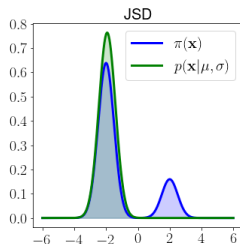
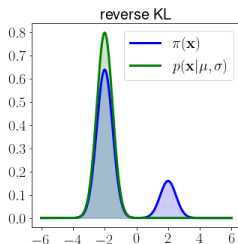
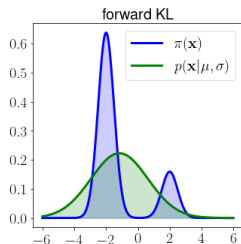
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Theoretical Results

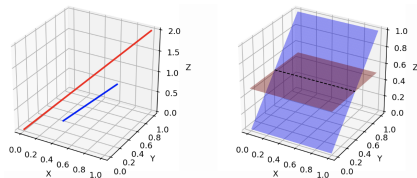
- ▶ The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_{\theta}(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ lives on a low-dimensional manifold.

Weng L. *From GAN to WGAN*, 2019

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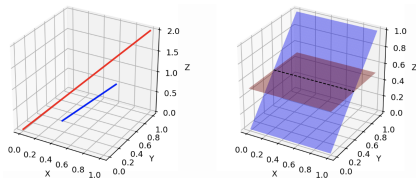


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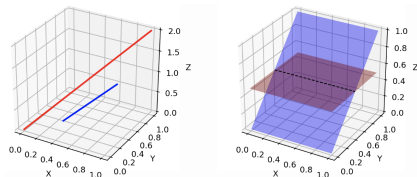
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- ▶ If $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!
- ▶ For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \parallel p_{\theta}) = D_{\text{KL}}(p_{\theta} \parallel p_{\text{data}}) = \infty, \quad D_{\text{JS}}(p_{\text{data}} \parallel p_{\theta}) = \log 2$$

Weng L. *From GAN to WGAN*, 2019

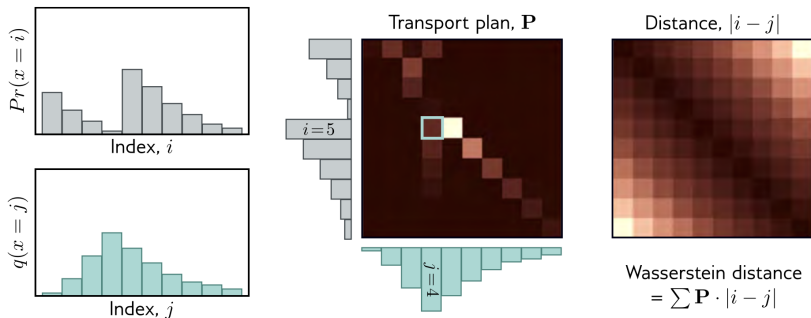
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Wasserstein Distance (Discrete)

Also known as the **Earth Mover's Distance**.

Optimal Transport Formulation

The minimum cost of moving and transforming a pile of “dirt” shaped like one probability distribution to match another.



Wasserstein Distance (Continuous)

$$W(\pi \| p) = \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| = \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2$$

- ▶ $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of “dirt” assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.

$$\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2); \quad \int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1).$$

- ▶ $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p .
- ▶ $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

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Wasserstein Metric

$$W_s(\pi, p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

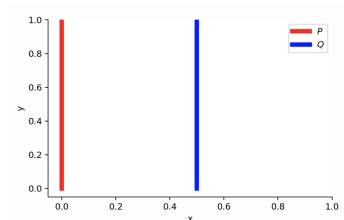
In our setting, $W(\pi \| p) = W_1(\pi, p)$, which is the transport cost formulation.

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1])$$

$$p_{\theta}(x, y) = (\theta, U[0, 1])$$



Weng L. *From GAN to WGAN*, 2019

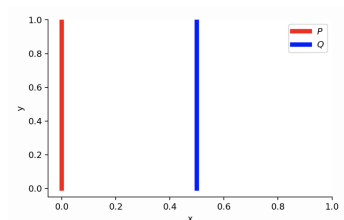
Arjovsky M., Chintala S., Bottou L. *Wasserstein GAN*, 2017

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- $\theta = 0$: Both distributions are identical.

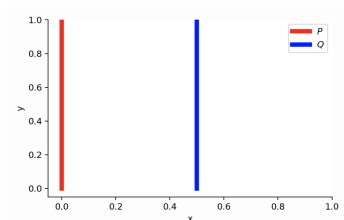
$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

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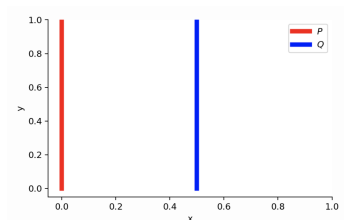
$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int_{U[0,1]} 1 \log \frac{1}{0} dy = \infty = D_{\text{KL}}(p_{\theta} \| p_{\text{data}})$$

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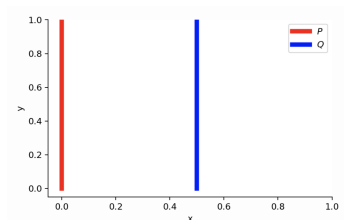
$$D_{\text{JS}}(p_{\text{data}} \| p_{\theta}) = \frac{1}{2} \left(\int_{U[0,1]} 1 \log \frac{1}{1/2} dy + \int_{U[0,1]} 1 \log \frac{1}{1/2} dy \right) = \log 2$$

Wasserstein Distance vs KL vs JSD

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Weng L. *From GAN to WGAN*, 2019

Arjovsky M., Chintala S., Bottou L. *Wasserstein GAN*, 2017

Wasserstein Distance vs KL vs JSD

Theorem 1

Let $\mathbf{G}_\theta(\mathbf{z})$ be (almost) any feedforward neural network, and $p(\mathbf{z})$ a prior over $\mathbf{z}$ such that $\mathbb{E}_{p(\mathbf{z})} \|\mathbf{z}\| < \infty$. Then $W(p_{\text{data}} \| p_\theta)$ is continuous everywhere and differentiable almost everywhere.

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Theorem 2

Let π be a distribution on a compact space $\mathcal{X}$ and let $\{p_t\}_{t=1}^\infty$ be a sequence of distributions on $\mathcal{X}$.

$$D_{\text{KL}}(\pi \| p_t) \rightarrow 0 \quad (\text{or } D_{\text{KL}}(p_t \| \pi) \rightarrow 0) \quad (1)$$

$$D_{\text{JS}}(\pi \| p_t) \rightarrow 0 \quad (2)$$

$$W(\pi \| p_t) \rightarrow 0 \quad (3)$$

As $t \rightarrow \infty$, $(1) \Rightarrow (2)$, and $(2) \Rightarrow (3)$. That is, convergence in Wasserstein distance is a weaker condition than convergence in JSD, which in turn is weaker than convergence in KL.

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Wasserstein GAN

Wasserstein Distance

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The infimum over all possible $\gamma \in \Gamma(\pi, p)$ is computationally intractable.

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Theorem (Kantorovich-Rubinstein Duality)

$$W(\pi \| p) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{\pi(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p(\mathbf{x})} f(\mathbf{x}) \right]$$

where $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is K -Lipschitz ($\|f\|_L \leq K$):

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq K \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}.$$

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We can thus estimate $W(\pi \| p)$ using only samples and a function f .

Wasserstein GAN

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\theta}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right]$$

- ▶ We must ensure that f is K -Lipschitz continuous.
- ▶ Let $f_{\phi}(\mathbf{x})$ be a feedforward neural network parameterized by ϕ .
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- ▶ Let $f_{\phi}(\mathbf{x})$ be a feedforward neural network parameterized by ϕ .
- ▶ If the weights ϕ are restricted to a compact set Φ , then f_{ϕ} is K -Lipschitz.
- ▶ Clamp weights within the box $\Phi = [-c, c]^d$ (e.g. $c = 0.01$) after each update.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\theta}) &= \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right] \geq \\ &\geq \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f_{\phi}(\mathbf{x}) \right] \end{aligned}$$

Wasserstein GAN

Standard GAN Objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})))$$

WGAN Objective

$$\min_{\theta} W(p_{\text{data}} \| p_{\theta}) \approx \min_{\theta} \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})) \right]$$

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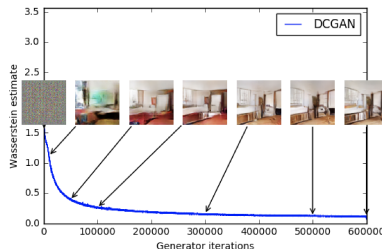
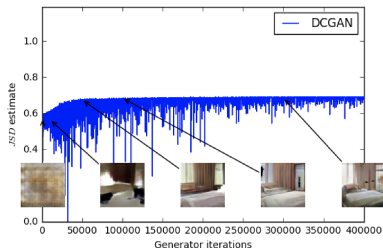
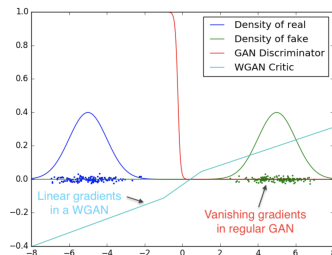
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- ▶ The discriminator D is replaced by function f : in WGAN, it is known as the **critic**, which is *not* a classifier.
- ▶ “*Weight clipping is a clearly terrible way to enforce a Lipschitz constraint.*”
 - ▶ If c is large, optimizing the critic is hard.
 - ▶ If c is small, gradients may vanish.

Wasserstein GAN

- ▶ WGAN provides nonzero gradients even if distributions' supports are disjoint.
- ▶ $D_{JS}(p_{\text{data}} \| p_{\theta})$ is poorly correlated with sample quality and remains near its maximum value $\log 2 \approx 0.69$.
- ▶ $W(p_{\text{data}} \| p_{\theta})$ is tightly correlated with quality.



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Evaluation of Likelihood-Free Models

Likelihood-Based Models

- ▶ **Train:** fit the model.
- ▶ **Validation:** tune hyperparameters.
- ▶ **Test:** assess generalization by reporting likelihood.

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Desirable Properties for Samples

- ▶ Sharpness



Low sharpness



High sharpness

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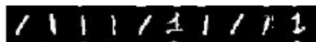


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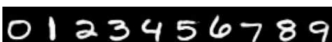


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- ▶ Diversity



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Theorem

If $\pi(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\pi, \boldsymbol{\Sigma}_\pi)$, $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, then

$$W_2^2(\pi \| p) = \|\boldsymbol{\mu}_\pi - \boldsymbol{\mu}_p\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_\pi + \boldsymbol{\Sigma}_p - 2 \left(\boldsymbol{\Sigma}_\pi^{1/2} \boldsymbol{\Sigma}_p \boldsymbol{\Sigma}_\pi^{1/2} \right)^{1/2} \right]$$

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Frechet Inception Distance

$$\text{FID}(p_{\text{data}}, p_\theta) = W_2^2(p_{\text{data}} \| p_\theta)$$

Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017

Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\mu_{\text{data}} - \mu_{\theta}\|^2 + \text{tr} \left[\Sigma_{\text{data}} + \Sigma_{\theta} - 2 \left(\Sigma_{\text{data}}^{1/2} \Sigma_{\theta} \Sigma_{\text{data}}^{1/2} \right)^{1/2} \right]$$

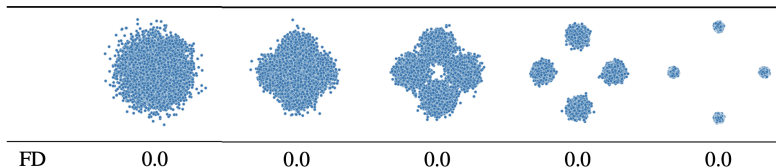
- ▶ FID is computed in the latent space $\mathbf{z}$.
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- ▶ μ_{data} , Σ_{data} and μ_{θ} , Σ_{θ} are statistics of latent representations for samples from $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$.

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$$\text{FID}(p(\mathbf{x}), \mathcal{N}(0, \mathbf{I}))$$



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Drawbacks

- ▶ Depends on the pretrained classification network.
- ▶ Uses the normality assumption.
- ▶ May not correlate with human evaluation.

Model	Model-A	Model-B
FID	21.40	18.42
FID _∞	20.16	17.19
Human rater preference	92.5%	6.9%

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Precision-Recall

CLIP Score

Human Eval

Precision-Recall

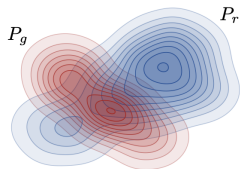
Desirable Properties for Samples

- ▶ **Sharpness**: generated samples should possess high visual quality.
- ▶ **Diversity**: their variation should match that in the training data.

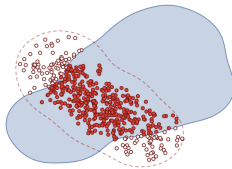
Precision-Recall

Desirable Properties for Samples

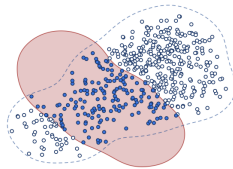
- ▶ **Sharpness**: generated samples should possess high visual quality.
- ▶ **Diversity**: their variation should match that in the training data.



(a) Example distributions



(b) Precision



(c) Recall

- ▶ **Precision** denotes the fraction of generated images that look realistic.
- ▶ **Recall** measures how well the generator covers the training data manifold.

Precision-Recall

- ▶ $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ – real samples;
- ▶ $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ – generated samples.

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Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

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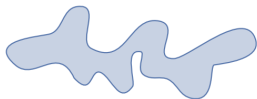
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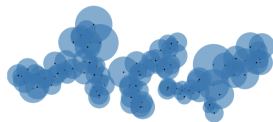
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(a) True manifold



(b) Approx. manifold

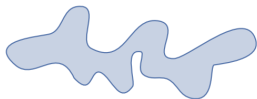
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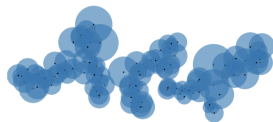
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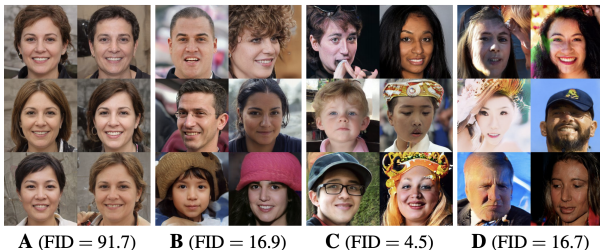
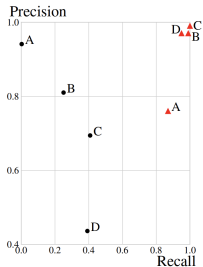
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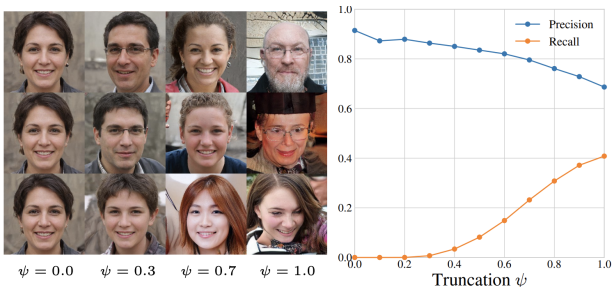
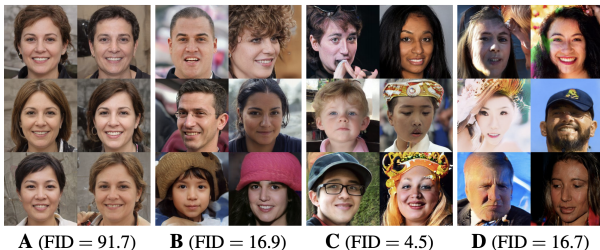
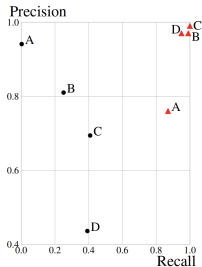
Embed the samples using a pretrained network (as in FID).

Precision-Recall



Kynkäänniemi T. et al. Improved precision and recall metric for assessing generative models, 2019

Precision-Recall



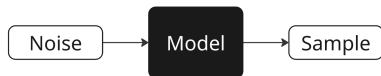
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Outline

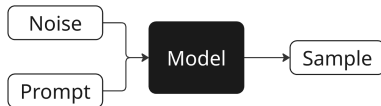
1. Generative Adversarial Networks (GAN)
2. Wasserstein Distance
3. Wasserstein GAN (WGAN)
4. Evaluation of Likelihood-Free Models
 - Frechet Inception Distance (FID)
 - Precision-Recall
 - CLIP Score**
 - Human Eval

CLIP Score

Unconditional Model

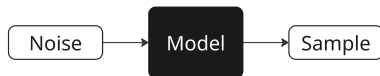


Conditional Model

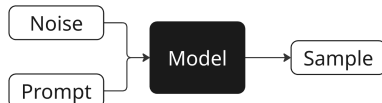


CLIP Score

Unconditional Model



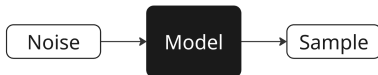
Conditional Model



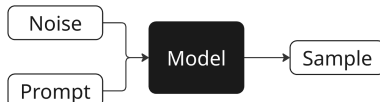
We need a way to measure not only the quality of the generated image, but also how well it's aligned with the prompt.

CLIP Score

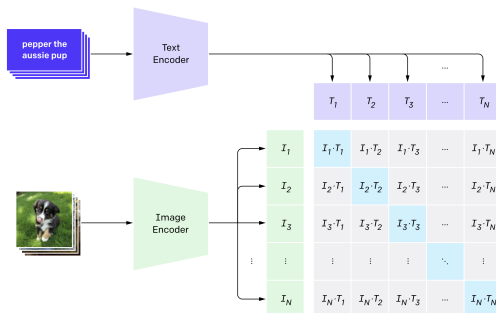
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Conditional Model



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Radford A. et al. *Learning transferable visual models from natural language supervision*, 2021

Outline

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Human Evaluation

- ▶ No automated metric is perfect.
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Аспект	Yandex ART 2.0	Mj 6.1	Mj 6	Ideogram	Recraft	Google Imagen3	Dall-E 3	FLUX	SBER Kandi3.1
Релевантность	0,59	0,58	0,63	0,45	0,51	0,50	0,50	0,54	0,75
Эстетика	0,49	0,55	0,55	0,51	0,51	0,61	0,61	0,54	0,59
Комплексность	0,44	0,73	0,70	0,68	0,76	0,75	0,75	0,71	0,74
Дефектность	0,69	0,57	0,68	0,55	0,59	0,63	0,63	0,50	0,75
Предпочтение	0,66	0,60	0,69	0,49	0,54	0,63	0,63	0,51	0,84

Summary

- ▶ GANs, in theory, optimize the Jensen-Shannon divergence.
- ▶ Wasserstein distance works in the case of disjoint data and model distributions (unlike the KL and JS divergences).
- ▶ Wasserstein GAN uses Kantorovich-Rubinstein duality to enable Monte Carlo estimation of the Wasserstein distance. It enforces the Lipschitz condition on the critic through weight clipping.
- ▶ FID is the most popular metric for evaluating implicit generative models.
- ▶ Precision-recall allows for choosing a model that balances sample quality and diversity.
- ▶ The CLIP score is widely used to measure text-to-image alignment.
- ▶ The gold standard for evaluating generated image quality is human assessment.